

aws

AWS Simple Storage Service S3



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Amazon Simple Storage Service (S3)
provides **object storage** through a web
interface.

S3 provides high availability, durability,
and **cost-effective storage** for backups,
big data analytics, and cloud
applications.



S3 Bucket

- A bucket in Amazon S3 is a container for **storing objects (files)**.
- Buckets are created in a specific AWS region and have **globally unique names**.
- Buckets are defined at the **region level**.
- You can control access to buckets using **bucket policies, ACLs, and IAM roles**.

Example:

- Bucket: **my-bucket-unique**



S3 Bucket

Amazon S3 > Buckets > Create bucket

Info

Create bucket

Buckets are containers for data stored in S3.

General configuration

AWS Region
Europe (Frankfurt) eu-central-1

Bucket name | [Info](#)

my-aws-bucket-unique

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership

Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced



S3 Objects (Files)

- Files stored in S3 are called **objects**.
- Each object consists of data, a **key** (file name, full **path**), metadata, and optionally tags.
- Objects are stored within buckets and identified using a unique key (like a file path).

Example:

- Bucket: **my-bucket-unique**
- Key: **AWS-logo-s3.png**
- S3 address:
 - `s3://my-bucket-unique/AWS-logo-s3.png`



S3 Features

- Max. Object Size is **5TB (5000GB)**
- If uploading more than 5GB, it must be used **multi-part upload**
- **Scalability:** Unlimited storage.
- **Durability:** 99.999999999% (11 9s) durability.
- **Security:** Encryption and access management.
- **Cost-effectiveness:** Multiple storage classes.



How to create S3 Bucket?

Amazon S3 > Buckets > Create bucket

Info

Create bucket

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Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced



How to create S3 Bucket?

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ Block *all* public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☒ Block public access to buckets and objects granted through *new* access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ Block public access to buckets and objects granted through *any* access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☒ Block public access to buckets and objects granted through *new* public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ Block public and cross-account access to buckets and objects through *any* public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☒ Disable

☐ Enable



How to create S3 Bucket?

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the **Storage** tab of the [Amazon S3 pricing page](#). [↗](#)

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#) [↗](#)

- ☐ Disable
- ☒ Enable

▼ Advanced settings

Object Lock

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#) [↗](#)

- ☒ Disable
- ☐ Enable
Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.

[i](#) Object Lock works only in versioned buckets. Enabling Object Lock automatically enables Versioning.



How to create S3 Bucket?

Amazon S3 > Buckets > my-aws-bucket-unique > Upload

Upload

Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 20.2 KB)

Remove

Add files

Add folder

All files and folders in this table will be uploaded.

Find by name

< 1 >

☐	Name	Folder	Type	Size
☐	AWS-logo-s3.png	-	image/png	20.2 KB

Destination

Info

Destination

[s3://my-aws-bucket-unique](#)

Destination details

Bucket settings that impact new objects stored in the specified destination.

Permissions

Grant public access and access to other AWS accounts.

Properties

Specify storage class, encryption settings, tags, and more.

Cancel

Upload

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Storage Classes

Amazon S3 offers storage classes to **optimize costs** based on **access patterns** and performance requirements.

- **S3 Standard:** For frequently accessed data.
- **S3 Intelligent-Tiering:** Automatically moves data to the most cost-efficient tier.
- **S3 Glacier Instant Retrieval:** Low-cost storage for infrequent access.
- **S3 Glacier Flexible Retrieval:** Archive data with retrieval times of minutes to hours.
- **S3 Glacier Deep Archive:** Lowest cost for long-term archival.



Storage Classes

S3 Lifecycle Rules

- Automatically manage object storage over time to optimize costs.
- **How:** Define rules to transition objects between storage classes or delete them after a set period.
- **Examples:**
 - Move objects from S3 Standard to S3 Glacier after 30 days.
 - Permanently delete objects older than 365 days.



S3 Event Notifications

- Trigger actions when specific events happen in your S3 bucket.
- **Supported Destinations:**
 - **Amazon SNS:** Send notifications.
 - **Amazon SQS:** Send messages to a queue.
 - **AWS Lambda:** Trigger serverless functions.
- **Examples:**
 - **Trigger a Lambda function when a new file is uploaded.**
 - **Notify a monitoring system when files are deleted.**



S3 Bucket Policy

- Define access permissions at the bucket level using JSON-based policies.
- **Granularity:** Controls access for IAM users, roles, or external accounts.
- **Principle of Least Privilege:** Grant the minimum necessary permissions.
- **Avoid Public Buckets:** Only make public when absolutely necessary, and monitor with AWS Trusted Advisor.
- **Use IAM Roles for Applications:** Assign IAM roles to applications instead of hardcoding credentials.
- **Enable Bucket Policy Logging:** Use AWS CloudTrail to track changes and access patterns.



S3 Bucket Policy Details

- **Version:** Defines the policy language version (always use "2012-10-17").
- **Statement:** Array of rules for access control. Each statement has:
 - **Effect:** Allow or Deny.
 - **Principal:** Who can access the bucket (IAM roles, users, AWS services).
 - **Action:** S3 actions like s3:GetObject or s3:PutObject.
 - **Resource:** The bucket or objects being accessed.
 - **Condition (optional):** Adds context-based rules like IP address restrictions.



S3 Bucket Policy Details

Edit bucket policy [Info](#)

Bucket policy

[Policy examples](#) [Policy generator](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN
 arn:aws:s3:::my-aws-bucket-unique

Policy

1		Edit statement
		Select a statement Select an existing statement in the policy or add a new statement. + Add new statement



S3 Bucket Policy Details

AWS Policy Generator

The AWS Policy Generator is a tool that enables you to create policies that control access to [Amazon Web Services \(AWS\)](#) products and resources. For more information about creating policies, see [key concepts in Using AWS Identity and Access Management](#). Here are [sample policies](#).

Step 1: Select Policy Type

A Policy is a container for permissions. The different types of policies you can create are an [IAM Policy](#), an [S3 Bucket Policy](#), an [SNS Topic Policy](#), a [VPC Endpoint Policy](#), and an [SQS Queue Policy](#).

Select Type of Policy S3 Bucket Policy ▼

Step 2: Add Statement(s)

A statement is the formal description of a single permission. See [a description of elements](#) that you can use in statements.

Effect ☒ Allow ☐ Deny

Principal

Use a comma to separate multiple values.

AWS Service Amazon S3 ▼ ☐ All Services ('*')

Use multiple statements to add permissions for more than one service.

Actions -- Select Actions -- ▾ ☐ All Actions ('*')

Amazon Resource Name (ARN)

ARN should follow the following format: `arn:aws:s3:::${BucketName}/${KeyName}`.
Use a comma to separate multiple values.

[Add Conditions \(Optional\)](#)

Add Statement

Step 3: Generate Policy

A *policy* is a document (written in the [Access Policy Language](#)) that acts as a container for one or more statements.

Add one or more statements above to generate a policy.



S3 Bucket Policy

IAM Policy

- **S3 Bucket Policy:**
 - Add a bucket policy to explicitly **allow the IAM user's principal (ARN)** to access the bucket.
 - This is crucial when you need to provide access across accounts or add additional restrictions (e.g., IP conditions).
- **IAM Policy (User or Role):**
 - Attach an IAM policy to the user (or to an IAM role the user assumes) that grants permissions to interact with the bucket.
 - This policy defines what actions the user can perform and on which bucket.



S3 Bucket Policy

```
{ } bucket-policy.json X
{ } bucket-policy.json > ...
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowListBucket",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::<AccountID>:user/<IAMUser>"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::<BucketName>/*"
    }
  ]
}
```



IAM Policy

```
{ } iam-policy.json X
{ } iam-policy.json > ...

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AccessSourceBucket",
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket",
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::<BucketName>/*"
    }
  ]
}
```



S3 Data Encryption

Protect your data with encryption at rest and in transit.

- **Server-side encryption (SSE):** SSE-S3, SSE-KMS, SSE-C.
- **Client-side encryption:** Data encrypted before uploading to S3.

Edit default encryption [Info](#)

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type | [Info](#)

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.](#)

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

[Cancel](#) [Save changes](#)



Access Management

Control who can access your data with flexible access policies.

- **AWS Identity and Access Management (IAM).**
- **Bucket policies and Access Control Lists (ACLs).**
- **AWS Organizations for cross-account access.**



Access Management

Access Control List

Access control list (ACL)

Edit

Grant basic read/write permissions to other AWS accounts. [Learn more](#)

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket.
Learn more about [using Amazon S3 Block Public Access](#)

The console displays combined access grants for duplicate grantees
To see the full list of ACLs, use the Amazon S3 REST API, AWS CLI, or AWS SDKs.

Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID:  744c069beddb1e2afa376ee94c5260f2c5358cd14497488c4c1729f2c870483a	List, Write	Read, Write
Everyone (public access) Group:  http://acs.amazonaws.com/groups/global/AllUsers	-	-
Authenticated users group (anyone with an AWS account) Group:  http://acs.amazonaws.com/groups/global/AuthenticatedUsers	-	-
S3 log delivery group Group:  http://acs.amazonaws.com/groups/s3/LogDelivery	-	-



Access Management Public Access

Object URL

 <https://my-aws-bucket-unique.s3.eu-central-1.amazonaws.com/AWS-logo-s3.png>

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" ?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>304PT6C9Y4QB3SDN</RequestId>
  <HostId>4WMQy/S9Jc7XL0ht7wii+w7pKbu00GDdqkJDYU6/eaQ+cbiWsRV0X1M14v0m5QccDmwqZDtcSc=</HostId>
</Error>
```

Edit Block public access (bucket settings) [Info](#)

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

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☐ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.

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☐ **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

[Cancel](#)

[Save changes](#)



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Access Management

Public Access

Amazon S3 > Buckets > my-aws-bucket-unique > AWS-logo-s3.png > Edit access contr...

Access control list (ACL)

Grant basic read/write permissions to AWS accounts. [Learn more](#)

Grantee	Objects	Object ACL
Object owner (your AWS account) Canonical ID: 744c069beddb1e2afa376ee94c5260f2c5358cd14497488c4c1729f2c870483a	☒ Read	☒ Read ☒ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☒ ⚠ Read	☐ Read ☐ Write
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ Read	☐ Read ☐ Write

⚠ When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access this object.
[Learn more](#)

☒ I understand the effects of these changes on this object.

Access for other AWS accounts

No other AWS accounts associated with the resource.

Add grantee

Specified objects

Name	Type	Version ID	Last modified	Size
AWS-logo-s3.png	png	-	March 5, 2025, 13:51:37 (UTC+01:00)	20.2 KB

Cancel

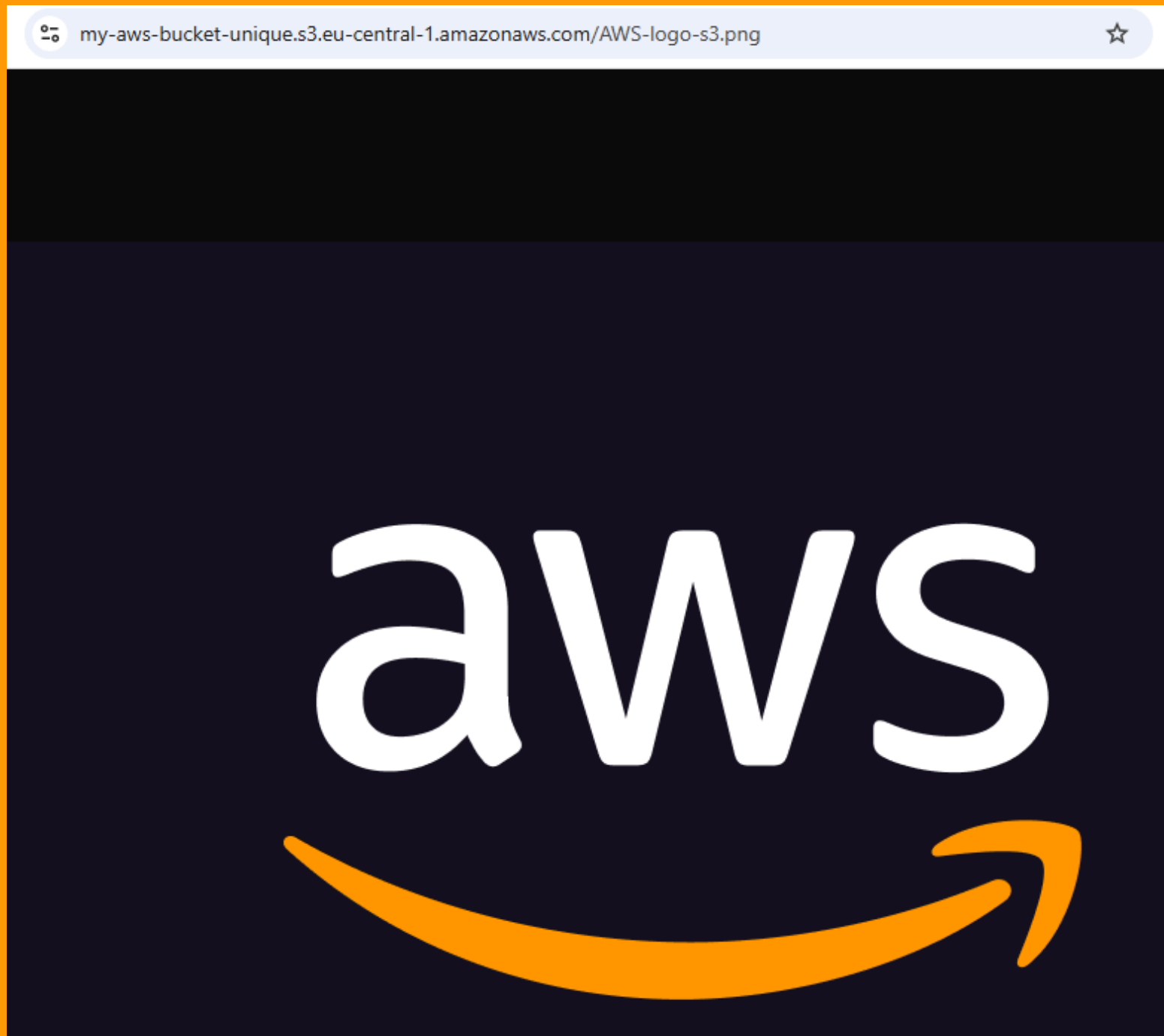
Save changes



Access Management Public Access

Object URL

<https://my-aws-bucket-unique.s3.eu-central-1.amazonaws.com/AWS-logo-s3.png>



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S3 Object Lock and Versioning

- **Object Lock:** Prevents data deletion or modification.
- **Versioning:** Maintains multiple versions of an object.

Bucket Versioning

[Edit](#)

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#) 

Bucket Versioning

Enabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#) 

Disabled



Static Website Hosting

- Host static websites directly from an S3 bucket.
- Configure bucket permissions for public access

The screenshot shows the AWS Management Console interface for editing static website hosting on a bucket named 'my-aws-bucket-unique'. The breadcrumb navigation at the top reads: 'Amazon S3 > Buckets > my-aws-bucket-unique > Edit static website hosting'. The main heading is 'Edit static website hosting' with an 'Info' link. Below this, the 'Static website hosting' section has two radio buttons: 'Disable' and 'Enable', with 'Enable' selected. The 'Hosting type' section has two radio buttons: 'Host a static website' (selected) and 'Redirect requests for an object'. Under 'Host a static website', there is a note: 'Use the bucket endpoint as the web address. [Learn more](#)'. Under 'Redirect requests for an object', there is a note: 'Redirect requests to another bucket or domain. [Learn more](#)'. A light blue information box contains a note: 'For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)'. The 'Index document' section has a text input field containing 'index.htm'. The 'Error document - optional' section has a text input field containing 'error.html'. The 'Redirection rules - optional' section has a note: 'Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)'. At the bottom, there is a table with one row, numbered '1' in the first column, and an empty second column.



Static Website Hosting

Static website hosting

[Edit](#)

Use this bucket to host a website or redirect requests. [Learn more](#)

ⓘ We recommend using AWS Amplify Hosting for static website hosting

[Create Amplify app](#)

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#)

S3 static website hosting
Enabled

Hosting type
Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://my-aws-bucket-unique.s3-website.eu-central-1.amazonaws.com>



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Static Website Hosting

Amazon S3 > Buckets > my-aws-bucket-unique > **index.html** > Edit access...

Edit access control list Info

Access control list (ACL)

Grant basic read/write permissions to AWS accounts. [Learn more](#)

Grantee	Objects	Object ACL
Object owner (your AWS account) Canonical ID: 744c069be ddb1e2afa376ee94c5260f2c 5358cd14497488c4c1729f2c 870483a	☒ Read	☒ Read ☒ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☒ Read	☐ Read ☐ Write
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ Read	☐ Read ☐ Write

When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access this object.

[Learn more](#)

☒ I understand the effects of these changes on this object.

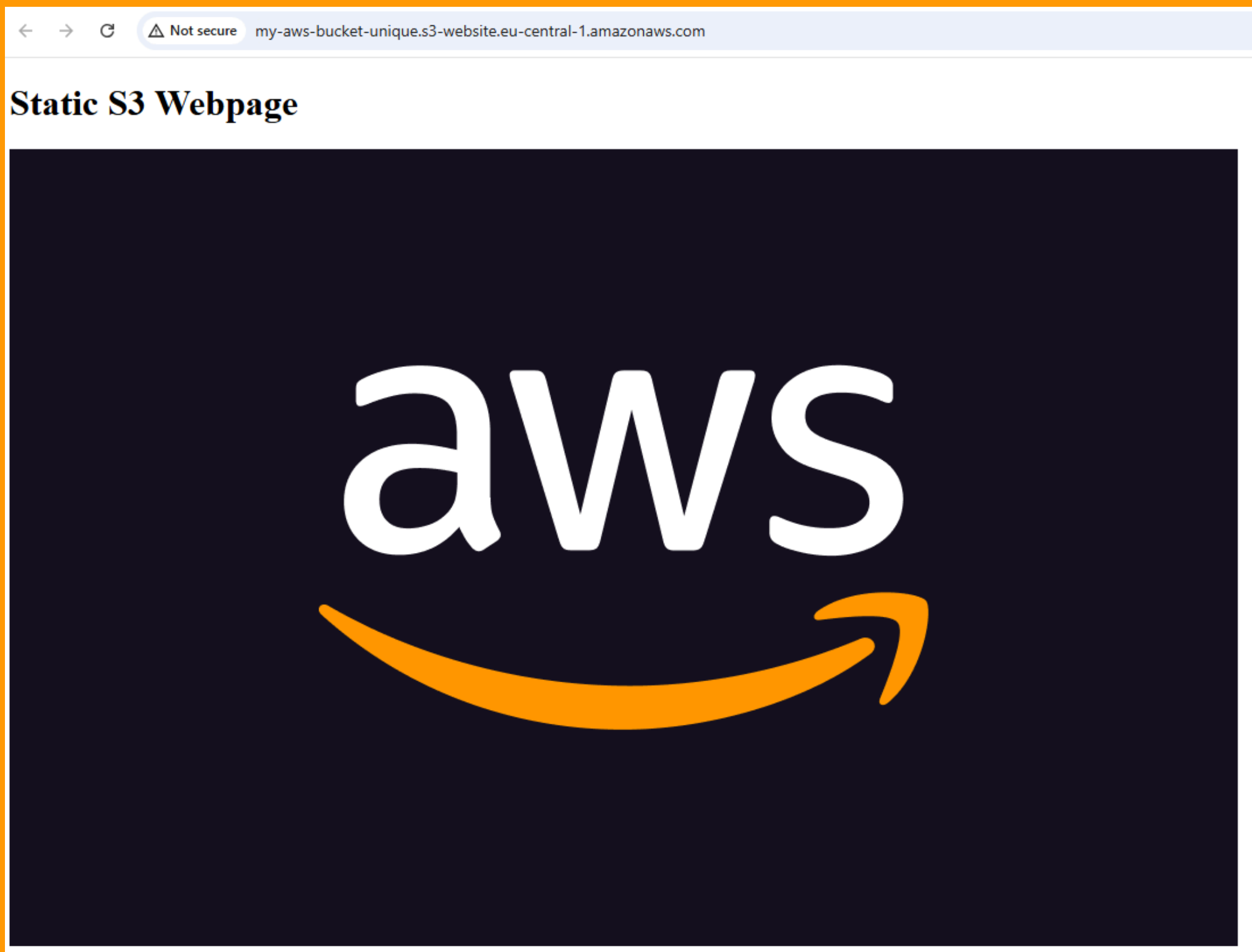
Access for other AWS accounts

No other AWS accounts associated with the resource.

[Add grantee](#)



Static Website Hosting



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S3 Access Points

S3 Access Point is a feature that provides a dedicated, customizable endpoint to simplify and control access to S3 buckets for specific applications or user groups

Amazon S3 > my-aws-bucket-unique > Create access point

Create access point [Info](#)

Amazon S3 Access Points simplify managing data access at scale for shared datasets in S3. Access points are named network endpoints that are attached to buckets that you can use to perform S3 object operations. [Learn more](#)

Properties

AWS Region
Europe (Frankfurt) eu-central-1

Access point name

Access point names must be unique within the account for this Region and comply with the [rules for access point naming](#).

Bucket name
my-aws-bucket-unique

Network origin
☒ Virtual private cloud (VPC)
No internet access. Requests are made over a specified VPC only.
☐ Internet

ⓘ The S3 console doesn't support accessing bucket resources using a virtual private cloud (VPC) access point. To access bucket resources from a VPC access point, use the AWS CLI, AWS SDK, or Amazon S3 REST API. [Learn more](#)

VPC ID

VPC ID must start with vpc-



user@aws:\$ #####

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Terraform, Ansible, DevOps, AI/ML
Why? Cause; More will unfold over time



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<https://github.com/omerbsezer>

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to help more people see it

user@aws:\$ #####



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